GAIA 2.O: SMART FORESTT MONITORING SYSTEM
STUDENTS REPORT

THIVYADHARSHINI (RAK SENSORS)

RAK Node Sensor Integration & System Implementation

System Overview

The system consists of multiple distributed RAK-based nodes deployed for environmental monitoring in forest conditions. Each node is configured to perform specific sensing functions, enabling distributed data acquisition across different environmental parameters. This modular node-based approach allows flexible deployment and scalability depending on monitoring requirements.

The implementation focuses on hardware-level sensor integration, stable system operation, and reliable communication with the LoRaWAN gateway under low-power constraints. The nodes are designed to function autonomously in remote environments with minimal maintenance.

Multi-Sensor Integration

Each node integrates a combination of WisBlock sensor modules to capture environmental, motion, and event-based data:

- **RAK1906 (BME680)** - Temperature, humidity, pressure, and gas sensing
- **RAK12033 (IIM42652)** - 6-axis accelerometer and gyroscope
- **RAK12020** - Ambient light intensity measurement
- **RAK12006 (PIR)** - Motion detection
- **RAK12015** - Vibration-based disturbance detection
- **RAK12500 (u-blox GNSS)** - Location tracking
- **RAK12023 + RAK12035** - Soil moisture and soil temperature sensing
- **RAK12005 + RAK12030** - Rain detection system
- **RAK18001** - Buzzer module for alert indication

The integration ensures compatibility across multiple modules operating simultaneously, with coordinated data acquisition and stable interfacing. Sensor outputs are validated to maintain consistency and accuracy across varying environmental conditions.

System Operation & Data Handling

The system operates using a hybrid approach combining periodic monitoring and event-driven detection:

•Scheduled data acquisition cycles for environmental parameters
•Immediate response to motion and vibration events
•Coordinated handling of multi-sensor data within a single operational cycle
•Structured data output for consistent interpretation and downstream usage

This approach ensures both continuous environmental tracking and real-time detection of critical events such as movement or disturbances.

Power Optimization & Low-Power Operation

Efficient power usage is critical for long-term deployment in remote forest environments.
Key operational strategies include:

•Utilization of deep sleep mode to minimize idle power consumption
•Event-based wake-up using motion and vibration triggers
•Time-based wake-up for periodic monitoring intervals
•Selective activation of sensors and modules to reduce unnecessary power usage
•Stable operation under battery-powered conditions

These mechanisms collectively enhance energy efficiency and extend system operational lifetime.

LoRaWAN Communication & Gateway Integration

The nodes use LoRaWAN communication (IN865 region) to enable long-range, low-power data transmission.

Key implementation aspects include:

•OTAA-based network activation using secure credentials
•Structured payload transmission for multi-sensor data
•Reliable uplink communication with the LoRaWAN gateway
•Integration with the gateway for end-to-end data transfer to the network server
•Verification of payload integrity and successful data reception

Gateway handling includes configuration, connectivity validation, and monitoring of communication reliability across nodes.

System Validation

Comprehensive testing and validation were performed to ensure system reliability:

•Verification of sensor outputs across different environmental conditions
•Validation of motion and vibration detection accuracy
•Testing of communication reliability between node and gateway
•Confirmation of stable operation during repeated sleep–wake cycles

These validation steps ensure consistent performance in real-world deployment scenarios.

Conclusion

The implemented RAK-based node system provides a scalable and energy-efficient solution for environmental monitoring in remote forest environments. Through effective sensor integration, optimized system operation, and reliable LoRaWAN communication, the system enables continuous data collection and responsive event detection, supporting long-term deployment with minimal maintenance.

KANMANNI (BMS REPORT)

Raspberry Pi 2W - Power System Design

Battery Selection: 2S $LiFePO_4$ (6 V, 6 Ah)

A 6 V, 6 Ah Lithium Iron Phosphate ($LiFePO_4$) battery pack (2S configuration using 32650 cells) is used to power the node. This battery is rechargeable and suitable for continuous operation of the Raspberry Pi with periodic activity from the RAK module.

Key Advantages:

- Stable voltage profile (7.2 V → ~5.5 V) → supports consistent converter input
- High safety and thermal stability → suitable for long-term deployment
- Long cycle life (2000+ cycles) → reliable for repeated usage
- Moderate energy capacity (~38 Wh)
- Low internal resistance → capable of supplying continuous load current

Justification:

Well suited for systems where the Raspberry Pi runs continuously, requiring a stable and rechargeable power source with sufficient energy capacity.

Power Conversion Requirements

A buck converter is required to step down the battery voltage to meet the Raspberry Pi input requirements.

- Input Voltage: 5.5 V – 7.2 V (2S $LiFePO_4$ battery range)
- Output Voltage: 5 V (regulated for Raspberry Pi)
- Peak Current: ≥ 2 A (to handle Pi + peripherals)
- Efficiency: > 85%
- Requirement: Stable output with minimal voltage drop during load variations

Selected Converter: MP1584 DC-DC 3A Buck Converter

The MP1584 buck converter module is selected based on the following:

- Wide input range (4.5 V – 28 V) → compatible with battery voltage range
- Adjustable output → configured to stable 5 V for Raspberry Pi
- Current capability (~3 A) → supports continuous load and transient spikes
- High efficiency (~90–95%)
- Compact size → suitable for embedded node design

Justification:

Provides **efficient and stable 5 V output** required for continuous Raspberry Pi operation while minimizing power losses.

Protection Element: 1.5 A Slow-Blow Fuse

A **1.5 A slow-blow fuse** is placed between the battery and the converter.

Key Features:
Allows temporary current spikes from Raspberry Pi without tripping
Protects against:
short circuits
converter faults
wiring failures

Ensures safe current limiting for the system

Conclusion

The combination of 2S $LiFePO_4$ battery, MP1584 buck converter, and 1.5 A slow-blow fuse provides a stable, efficient, and protected power solution for the node. This design ensures reliable continuous operation of the Raspberry Pi while accommodating intermittent activity from the RAK module.

Raspberry Pi 4 Model B – Power System Design

Battery Selection: 2S $LiFePO_4$ (6 V, 6 Ah)

A 6 V, 6 Ah Lithium Iron Phosphate ($LiFePO_4$) battery pack (2S configuration using 32650 cells) is selected to power the node. The battery supports continuous operation of the Raspberry Pi 4 Model B, while the RAK module operates intermittently and is powered through the Pi.

Key Advantages:

- Stable discharge profile (7.2 V → ~5.5 V) → ensures consistent converter input
- High safety and thermal stability → suitable for long-term outdoor deployment
- Long cycle life (2000+ cycles) → reliable for repeated usage
- Moderate energy capacity (~38 Wh)
- Low internal resistance → supports continuous load and startup current spikes

Justification:
Since the Raspberry Pi operates continuously, the system requires a rechargeable battery with stable voltage and sufficient energy capacity. The $LiFePO_4$ chemistry provides a reliable and safe solution for sustained operation with minimal voltage fluctuation.

Power Conversion Requirements

A buck converter is required to step down the battery voltage to a regulated 5 V supply for the Raspberry Pi.

Specifications:

- Input Voltage: 5.5 V – 7.2 V (battery range)
- Output Voltage: 5 V (regulated for Raspberry Pi)
- Peak Current: ≥ 2.5 A
- Efficiency: > 85%
- Requirement: Stable output under continuous load and transient spikes

Selected Converter: MP1584 DC-DC Buck Converter

The MP1584 DC-DC Buck Converter Module is selected based on the following:

Features:

- Wide input range (4.5 V – 28 V) → compatible with battery
- Adjustable output → set to ~5.1 V for stable Pi operation
- High efficiency (~90–95%)
- Compact and suitable for embedded design

Justification:

The converter provides efficient step-down conversion for continuous Raspberry Pi operation.

Protection Element: 3.15 A Slow-Blow Fuse

A 1.5 A slow-blow fuse is placed between the battery and the converter.

Key Features:
Allows temporary current spikes from Raspberry Pi without tripping
Protects against:
short circuits
converter faults
wiring failures

Ensures safe current limiting for the system

Conclusion

The combination of a 2S $LiFePO_4$ battery, MP1584 DC-DC Buck Converter Module, and 1.5 A slow-blow fuse provides a compact and efficient power system for the node.

This design ensures:

- continuous operation of the Raspberry Pi
- controlled intermittent operation of the RAK module
- safe and reliable power delivery under varying load conditions

The system is suitable for embedded and field-based monitoring applications requiring continuous processing with periodic communication.

RAK Node Power System Design

Battery Selection: Li-$SOCl_2$

A 3.6 V, 2.6 Ah Lithium Thionyl Chloride (Li-$SOCl_2$) battery from Saft is used to power the RAK node. Although non-rechargeable, it is highly suitable for remote forest deployment due to its long operational life and reliability.

Key Advantages:

- Very low self-discharge → enables multi-year operation
- High energy density → compact and lightweight design
- Stable output voltage (~3.6 V) → consistent performance
- Wide temperature tolerance → suitable for harsh environments
- Maintenance-free → ideal for inaccessible locations

Justification: Best suited for low-power, long-duration IoT applications where battery replacement is difficult.

Power Conversion Requirements

A buck converter is required to regulate the battery output for the RAK19007 WisBlock Base Board.

- **Input Voltage:** 5.4 V – 7.6 V (2S battery pack)
- **Output Voltage:** 3.7 V – 4.2 V
- **Peak Current:** ≥ 300 mA
- **Efficiency:** > 85%
- **Requirement:** Stable, low-ripple output

3. Selected Converter: XL4015

The XL4015 is selected based on the following:

- Wide input range (4 V – 38 V) → supports battery range
- Adjustable output → set to ~3.9–4.3 V
- High current capacity (5 A) → large safety margin
- High efficiency (~90%) → reduces power loss
- Built-in current limiting → basic protection

Conclusion

The combination of Li-$SOCl_2$ battery and XL4015 converter provides a reliable, efficient, and low-maintenance power solution for forest-deployed RAK nodes, ensuring long life and stable operation.

VISHWAR (FIRE DETECTION ML MODEL)

Technical Report: Gaia Smart Forest Monitoring System

Introduction

The *Gaia Smart Forest Monitoring System* is an intelligent solution designed to monitor forest environments using edge AI technology. The system aims to detect potential fire hazards in real time by analyzing visual data captured through AI-enabled cameras and processed on embedded devices.

Objective

The main objectives of my work were:

- To develop a **fire and smoke detection machine learning model**
- To classify fire-related scenarios into multiple categories
- To deploy the trained model on an embedded AI system
- To automate periodic environmental monitoring

Hardware Components Used

- **Raspberry Pi Zero 2 W** – Used for edge deployment and processing
- **Sony IMX AI Camera** – Used for capturing and processing visual data

Dataset and Class Distribution

The dataset was collected from **GitHub sources** and categorized into four classes:

1. Fire only
2. Smoke only
3. Fire and smoke
4. No fire (normal condition)

Dataset Split Ratio

For training and evaluation, the dataset was divided as:

- **70% Training Data**
- **30% Testing Data**

Why 70:30 Ratio?

A **70:30 split** is widely used because:

- It provides sufficient data for the model to learn patterns
- It keeps a good portion for unbiased testing
- It avoids overfitting compared to higher training ratios like 90:10

For your dataset size and 4-class classification problem, **70:30 is an appropriate and balanced choice**.

Model Training

- The model was trained using the prepared dataset
- Total training duration: **120 epochs**
- Achieved accuracy: **80.69%**
- The model successfully learned to classify fire and smoke conditions

Model Format and Conversion

- The trained model was saved as: **best.pt** (PyTorch format)
- It was converted into **IMX format** (Sony IMX-compatible binary format for edge AI cameras)

Format Explanation

- **.pt (PyTorch format):** Used during training and development
- **IMX format:** Optimized format required for deployment on Sony IMX AI cameras for efficient edge inference

Deployment

- The converted IMX model was deployed on the **Raspberry Pi Zero 2 W**
- The AI camera was interfaced with the Raspberry Pi for real-time data capture and processing

System Functionality

The system was programmed with the following workflow:

- The camera activates **once every 15 minutes**
- It captures a **short video clip**
- The captured clip is stored locally on the Raspberry Pi
- The deployed model analyzes the data for fire/smoke detection

Implementation Overview

- Dataset collection and preprocessing
- Multi-class ML model training
- Model optimization and format conversion
- Automated monitoring system development

Learning Outcomes

From this project, I gained experience in:

- Machine learning model development and evaluation
- Image classification techniques
- Working with PyTorch models and training pipelines
- Model conversion for embedded AI deployment
- Edge computing using Raspberry Pi
- Integration of AI hardware and software systems
- Automation in real-time monitoring applications

Conclusion

The Gaya Smart Fire Detection System demonstrates an efficient **edge AI-based monitoring solution**. With an accuracy of **80.69%**, the system is capable of detecting fire and smoke across multiple conditions. The use of a **70:30 dataset split** ensured balanced training and evaluation. This system can be further enhanced by:

- Increasing dataset size
- Improving model accuracy
- Adding real-time alert systems (SMS/IoT)
- Integrating cloud-based monitoring